

Lab1 Report

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Tasks

Create a program to decrypt the secret number with LC-3 machine code.

The secret number is placed in the register R0. The decryption process is to bitwise XOR the secret number with a secret key, which is determined by student ID (which is 01111111 or x7F in this case), and put the result in register R3.

The Core Algorithm

The algorithm includes two parts:

1. Input the secret key into register R2.
2. Do bitwise XOR between R0 and R2, put the result into register R3.

Part 1

As x7F is too large to fit into imm5 (used by ADD), it needs to be input by 3 steps:

1. Put x7 into register R2.
2. Shift them left by 4 bits.
3. Put xF into register R2.

The assembly code is as below.

```
AND R2, R2, x0
ADD R2, R2, x7
ADD R2, R2, R2
ADD R2, R2, R2
ADD R2, R2, R2
ADD R2, R2, R2
ADD R2, R2, xF
```

After translating to machine code:

```
0101 010 010 1 00000
0001 010 010 1 00111
0001 010 010 0 00 010
0001 010 010 0 00 010
0001 010 010 0 00 010
0001 010 010 0 00 010
0001 010 010 1 01111
```

Part 2

A way to realize bitwise XOR A with B is:

1. $A + B = C$.

2. $(A \text{ AND } B) * 2 = \text{CARRY}$, CARRY is the carry bits in A+B.

3. $C - \text{CARRY} = A \text{ XOR } B$

The assembly code is as below:

```
ADD R3,R0,R2
;put A+B in R3
AND R1,R0,R2
ADD R1,R1,R1
;put CARRY in R1
NOT R1,R1
ADD R1,R1,x1
ADD R3,R1,R3
;put R3-R1 in R3
```

After translating to machine code:

```
0001 011 000 0 00 010
0101 001 000 0 00 010
0001 001 001 0 00 001
1001 001 001 111111
0001 001 001 1 00001
0001 011 001 0 00 011
```

Run Results

Input 1: xFFFF

Output 1: xFF80

Input 2: x0000

Output 2: x007F

Input 3: xFF80

Output 3: xFFFF

Input 4: x1020

Output 4: x105F

Input 5: x2024

Output 5: x205B

All 5 tests passed, no bug found.

Suggestions

The bitwise XOR can also be realised by logical operations, as

$A \text{ XOR } B = (A \text{ AND } B) \text{ NOR } (A \text{ NOR } B)$